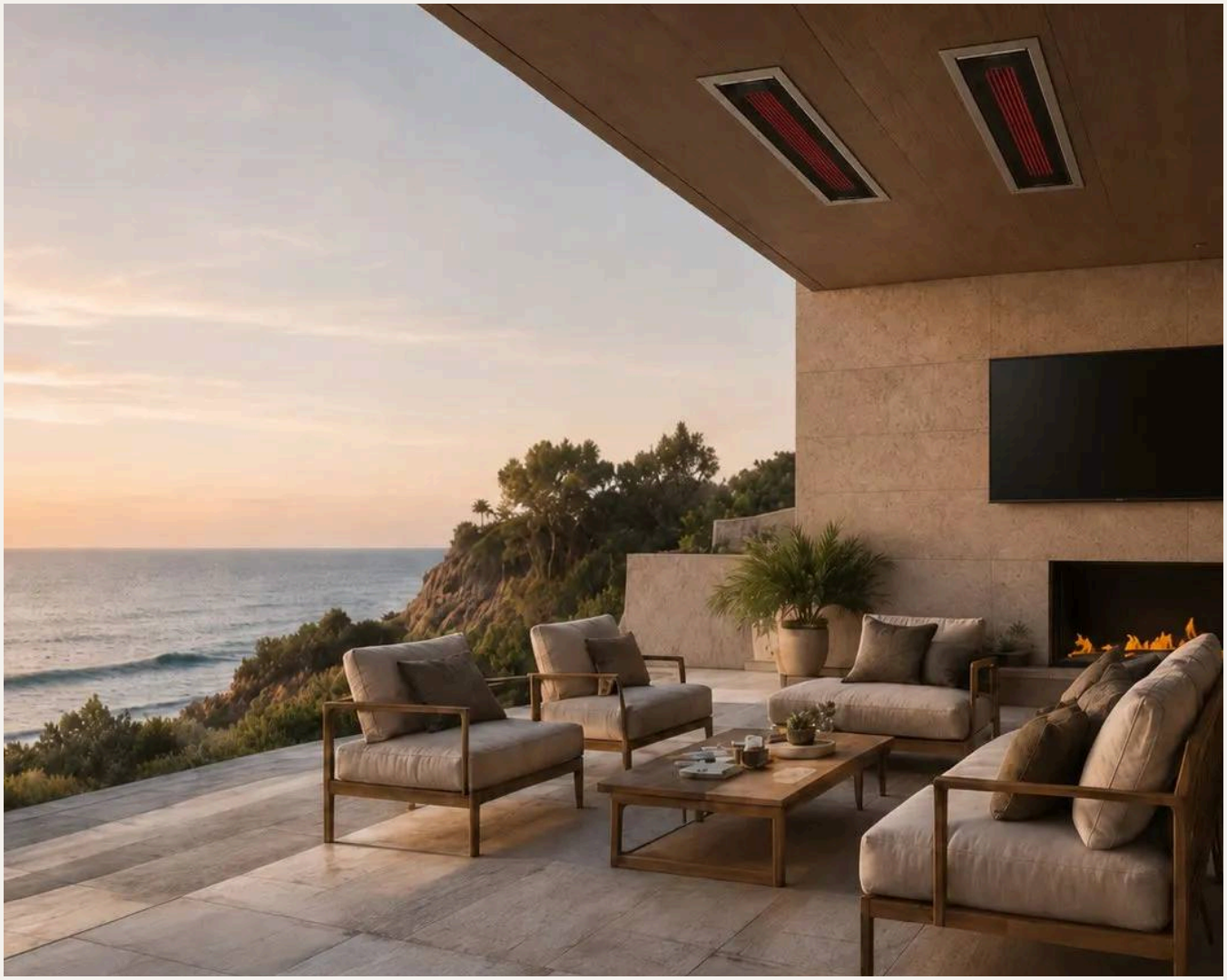
BROMIC
HEATING

RESIDENTIAL HOMEOWNER

The outdoor heating planning guide: homeowner edition

Plan patio heating in nine steps: measure, choose fuel and style, size it, plan power, controls and code, with running-cost maths and checklists.



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BROMIC HEATING
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The outdoor heating planning guide

Homeowner edition. Designed from the start, not added later.

Plan the heat in the same order you plan the patio: confirm the space counts as outdoors, measure it, choose electric or gas, choose a style, size it, then plan the power, the controls and the code before the ceiling closes. Decide early and the wiring or gas line goes in once, where you want it.

Only 23% of homeowners prioritize outdoor heating at the planning stage[1]. The rest add it later, with opened ceilings, surface conduit and heaters wherever the wiring happened to be.



Ocean-view outdoor living room with recessed heaters

The nine steps



STEP	DECISION	YOU'LL NEED
1	Is it outdoors?	Wall and roof layout
2	Measure	Tape measure, ceiling height
3	Choose the fuel	What is at the house; your utility bill
4	Choose style and mounting	A look you like; where heat can mount
5	Size it	Floor area; the free calculator
6	Plan the power or gas	An electrician or gas fitter
7	Choose the controls	How you want to switch heat on
8	Clearances and code	Your local building department
9	Book a design consult	Photos or plans

Each step ends with its checklist. Together they are the brief for your designer.

Step 1. Is it outdoors?



A space counts as outdoor when at least 30% of its perimeter is permanently open[3]. Most patios, porches, pergolas, loggias and covered terraces qualify. Closing it in further with glass or screens changes the design: talk to a designer first.

A roof helps. Warm air trapped under a cover forms a "heat cloud" that adds a gentle boost to the radiant heat[3].

[[DIAGRAM: Plan-and-section sketches of qualifying spaces: open courtyard with one ground-level opening; covered area with two open sides; three walls with 30% or more of the perimeter open; the heat cloud as a soft shaded zone under the cover. Rule per the 2021 guide[3].]]

- ☐ At least 30% of the perimeter permanently open[3]
- ☐ Open, partly covered or covered: _____
- ☐ Any plans to add glass, screens or blinds: _____

Step 2. Measure



- ☐ Length × width of the area you want warm = _____ ft² (measure over the seating, not the whole deck)
- ☐ Ceiling or beam height _____ ft; ceiling material _____
- ☐ Windy side _____; near water? Y / N
- ☐ Months and times of use _____

Step 3. Choose the fuel

Why radiant

Outdoors, warm air rises and blows away, so forced air and thermostats cannot hold an open patio at a set temperature[3]. Radiant (infrared) heat warms people, furniture and floors directly, like sun on your skin, so it keeps working in a light breeze. Heat you can feel, not see.

WAVELENGTH	RADIANT SHARE	WARM-UP	IN WIND
Short wave	about 90%	Instant	Warms people directly
Medium wave	about 60%	about 30 seconds	Limited sensitivity
Long wave	about 40%	about 5 minutes	Most affected

Source: Bromic Planning Guide 2021[3]. A fire pit is a focal point, not a heating plan.

Electric or gas

	ELECTRIC	GAS (NATURAL GAS OR PROPANE)
Suits	Covered patios, pergolas, loggias; ceilings, walls, poles	Larger, more open areas
Install	Hard-wired by a licensed electrician	Built-in: licensed gas fitter, plus a 110 V supply[10][11]. Freestanding: cylinder or quick-connect line[12]
Control	On/off or dimming, zones, schedules, app	On/off, wall switch, schedules
Coverage per heater	65–160 ft²	160–215 ft²

What it costs to run

- **Electric:** cost per hour = kW × $ per kWh
- **Natural gas:** cost per hour = Btu per hour ÷ 100,000 × $ per therm
- **Propane:** cost per hour = Btu per hour ÷ 91,500[23] × $ per gallon

Rates used, US averages: electricity 13.54¢/kWh (EIA, commercial, May 2026)[18]; natural gas $0.87 per therm (EIA, commercial, January–May 2026 average)[19]; propane $2.674 per gallon (EIA, residential, week of 30 March 2026)[20]. Home electricity and gas are billed at residential rates: put the rate from your own bill into the formula.

HEATER AT FULL OUTPUT	ENERGY PER HOUR	COST PER HOUR
Tungsten Electric 2000 / 3000 W	2.0 / 3.0 kWh	$0.27 / $0.41
Tungsten Electric 4000 / 6000 W	4.0 / 6.0 kWh	$0.54 / $0.81
Platinum Electric 2300 / 3400 / 4500 W	2.3 / 3.4 / 4.5 kWh	$0.31 / $0.46 / $0.61
Eclipse Electric 2900 W	2.9 kWh	$0.39
Platinum Gas 23,600 Btu, natural gas / propane	0.24 therm / 0.26 gal	$0.21 / $0.69
Tungsten Freestanding Gas 38,000 Btu, natural gas / propane	0.38 therm / 0.42 gal	$0.33 / $1.11

Dimming, or running an Eclipse at 33% or 66%[7], uses less. These replace the 2021 guide's figures, which used 2021 prices.

- ☐ Electric / natural gas / propane: _____
- ☐ Your rate from your bill: _____ per kWh, therm or gallon

Tungsten freestanding gas heater in a stone courtyard at dusk

Step 4. Choose style and mounting

FAMILY	LOOK	MOUNTING	FINISHES
Tungsten Smart-Heat Electric	Slim, architectural	Recessed, ceiling, wall	2000/3000 W Black; 4000/6000 W Black or White[4]
Platinum Smart-Heat Electric	Linear heater, Classic finish range	Wall, pole or recessed; 8 in pole included[5]	Black or White
Platinum Smart-Heat Electric Marine	316 marine-grade stainless for coastal homes	Wall, pole or recessed	Stainless[6]
Eclipse Smart-Heat Electric	Round pendant: heat and dimmable rim light in one	24 in twin pole standard; 8, 24 and 48 in straight, curved and wall-mount poles[7]	Black
Eclipse Smart-Heat Freestanding	The Eclipse on a weighted base with wheels and a curved arm	Freestanding	Black[2]
Cobalt Smart-Heat Electric	Slim linear heater	Wall or ceiling	Silver[9]
Platinum Smart-Heat Gas	Built-in gas, two sizes	Wall or ceiling; brackets included	Black or Stainless[10]
Tungsten Smart-Heat Gas	Built-in gas, two sizes; pivot arm ±25°	Wall or ceiling	Matte black[11]

FAMILY	LOOK	MOUNTING	FINISHES
Tungsten Smart-Heat Freestanding Gas	Tilting head, wheels, anti-tilt design	Freestanding	Black[2]
Tungsten Smart-Heat Electric, Coastal collection (coming soon)	316 marine-grade stainless; Signature Diamond Pattern grille	Recessed (low clearance), ceiling, wall, single-post	Black or White[2]
Tungsten Smart-Heat Electric, Select collection (coming soon)	Expanded fascia options	Recessed (low clearance), ceiling, wall, single-post	Black, White or Graphite[2]

A flush, recessed look needs a recess kit and must be decided before the ceiling is built.

- ☐ Family and finish: _____
- ☐ Mount: recessed / ceiling / wall / pole / freestanding
- ☐ Recess kit or longer pole needed? Y / N

Ceiling-mounted black heater over a candlelit lounge

Step 5. Size it



Heaters needed = area ÷ (coverage per heater × cover factor × climate factor), rounded up. A colder or more open site needs more heat for the same area[3]; the two factors put that into the sum:

COVER	FACTOR	CLIMATE (EVENINGS YOU HEAT)	FACTOR
Open (no roof)	0.85	Mild (above about 50°F)	1.0
Partly covered (pergola, umbrella, part roof)	1.0	Cool (about 40–50°F)	0.85
Covered (solid roof)	1.1	Cold (often below about 40°F)	0.7

Not sure? Use partly covered and cool: 1.0 × 0.85 = 0.85. Measure the seating, not the whole deck; if you know the seats rather than the area, allow 20 ft² a seat.

Freestanding heaters (Tungsten Freestanding Gas, Eclipse Freestanding) serve no more than four seats each, because the guests themselves block the radiant heat. Count both ways and take the larger: area ÷ (coverage × factors), or seats ÷ 4.

HEATER	COVERAGE PER HEATER (APPROX.)
Tungsten Electric 2000 / 3000 / 4000 / 6000 W	65 / 100 / 130 / 160 ft²[4]
Platinum Electric and Marine 2300 / 3400 / 4500 W	65 / 108 / 143 ft²[56]
Eclipse Electric and Eclipse Freestanding, 2900 W	121 ft²[2]
Cobalt Electric 4000 / 6000 W	130 / 160 ft²[9]

HEATER	COVERAGE PER HEATER (APPROX.)
Platinum Gas 23,600 / 39,800 Btu	160 / 215 ft²[10]
Tungsten Gas built-in, smaller / larger model	160 / 215 ft²[11]
Tungsten Freestanding Gas, 38,000 Btu	215 ft²[2]
Tungsten Electric Coastal and Select (coming soon)	64–144 ft² by model[2]

Coverage is approximate and varies with mounting height, wind and installation[2]. Product-page values read on 29 October 2026.

The free How many heaters? calculator does this for you with the same cover and climate factors.

Worked example: a 20 × 12 ft covered patio

1. **Area:** 20 × 12 = 240 ft². Covered, with one long side and one end open: well over 30% open[3].
2. **Factors:** covered 1.1 × cool evenings 0.85 = 0.935.
3. **Tungsten Electric 4000 W, recessed:** 240 ÷ (130 × 0.935) = 240 ÷ 121.6 = 2.0, so **two heaters**.
4. **Or Eclipse pendants over the table:** 240 ÷ (121 × 0.935) = 240 ÷ 113.1 = 2.1, so **three Eclipse**.
5. **Controls:** two Tungsten 4000 W heaters need two controller channels (Step 7): an Affinity Smart-Heat 2-channel Dimmer or On/Off controller, a wall switch and the app. Three Eclipse need an Affinity Eclipse controller channel each: a 5-channel controller, or three 1-channel.
6. **Running cost:** 2 × 4 kW × 3 hours × $0.1354 = about $3.25 a night at the May 2026 commercial average[18]. Use your own rate.

DIAGRAM: Plan of the 20 × 12 ft patio: two recessed Tungsten heaters along the long axis, or three Eclipse pendants over the table, with heat-bloom circles and the open sides marked.

- [ ] Area ÷ (coverage × cover × climate) = _____, rounded up = _____ heaters (freestanding: at least seats ÷ 4)
- [ ] Calculator run; result saved

Step 6. Plan the power or gas



Electric heaters are hard-wired. Your electrician sizes each circuit from the heater's current draw:

HEATER	AMPS AT 208 V	AMPS AT 240 V
Platinum Electric 2300 / 3400 / 4500 W	11.1 / 16.3 / 21.6	9.6 / 14.2 / 18.8[5]
Eclipse Electric 2900 W	13.9	12.1[7]
Eclipse Freestanding 2900 W	—	12.1[8]
Cobalt Electric 4000 / 6000 W	18.75 / 28.4	16.6 / 25[9]
Tungsten Electric 2000 / 3000 / 4000 / 6000 W	9.6 / 14.4 / 19.2 / 28.8	8.3 / 12.5 / 16.7 / 25.0

Tungsten Electric amps are calculated as watts ÷ volts; the product page does not list them.

Built-in gas heaters need a gas line from a licensed gas fitter and a 110 V, 2 A electrical supply[10][11]. The Tungsten Freestanding Gas runs on a 20 lb propane cylinder or a 10 ft natural-gas quick-connect line[12].

- ☐ Panel capacity checked by a licensed electrician
- ☐ Circuit per heater or group; junction-box positions marked
- ☐ Gas line and shut-off located (gas)
- ☐ Rough-in done before the ceiling closes

Step 7. Choose the controls (required)

A heater without a control is not a finished system. Every plan needs all four parts:

PART	WHAT IT DOES	CAPACITY
Affinity Smart-Heat On/Off controller	Switches heaters; electric or gas	1, 2 or 5 channels; 6 kW per channel at 208–277 V; up to 10 gas heaters per channel[13]
Affinity Smart-Heat Dimmer controller	Sets heat level in 1% steps; Tungsten and Platinum Electric	1, 2 or 5 channels; 6 kW per channel at 208–277 V[14]
Affinity Smart-Heat Eclipse controller	Heat steps and rim-light dimming for Eclipse	1 or 5 channels; one Eclipse per channel[15]
Manual wall switch	On/off at the door	Rated 20 A, 120–277 V[16]
Bromic Smart-Heat app	Zones, schedules, timers	Works through a compatible Affinity controller

Heaters per 6 kW channel: Tungsten 2000 W: 3 · 3000 W: 2 · 4000 or 6000 W: 1 · Platinum 2300 W: 2 · 3400 or 4500 W: 1 · Cobalt: 1 · Eclipse: 1 (on an Eclipse controller). Arithmetic on the 6 kW rating; the circuit can set a lower limit.

The Eclipse Freestanding comes with an Eclipse Dimmer Controller and remote[8]. Built-in gas switches on and off only; it does not dim[10].

Wi-Fi and app features require compatible Affinity Smart-Heat Controls, sold separately. The app also works with Amazon Alexa, Google Assistant, Control4, Crestron and Lutron; compatibility varies and may need extra hardware or setup[2].

- ☐ Controller type and channels: _____
- ☐ Wall-switch location: _____
- ☐ Wi-Fi reaches the controller? Y / N
- ☐ Smart-home system to connect: _____

Hand holding a tablet controlling heat at an outdoor table

Step 8. Clearances and code

HEATER	MINIMUM MOUNTING HEIGHT
Tungsten, Platinum and Platinum Marine Electric	Wall 72 in; ceiling 96 in[4][5][6]
Cobalt Electric	Wall 6 ft; ceiling 8 ft[9]
Eclipse Electric	96 in[7]
Eclipse Freestanding	79 in clearance[8]
Platinum Gas and Tungsten Gas built-in	96 in[10][11]

Clearances to the ceiling and to anything combustible are set by each model's installation manual. A heat deflector reduces the space a gas heater needs above it[3]. In the US, electric heaters are installed to ANSI/NFPA 70[21] and gas heaters to ANSI Z223.1/NFPA 54[22].

This is general guidance, not a code ruling. Confirm requirements with your local building authority, and have heaters installed by a licensed electrician or gas fitter to the manufacturer's instructions.

Warranty

PRODUCT	WARRANTY
Tungsten Electric	2 years heater; 7 years elements[4]

PRODUCT	WARRANTY
Platinum Electric	3 years[5]
Platinum Electric Marine	5 years on land; 3 years on vessels[6]
Eclipse Electric and Eclipse Freestanding	3 years[7][8]
Cobalt Electric	1 year parts; 7 years elements[9]
Platinum Gas and Tungsten Gas built-in	2 years, excluding hoses and igniters[10][11]
Tungsten Freestanding Gas	1 year (propane model)[12]
Affinity Smart-Heat Controls	1 year[13]

- [ ] Mount heights meet the minimums
- [ ] Clearances from each installation manual
- [ ] Permit and inspection confirmed with the building department

Eclipse pendants over a poolside loggia at blue hour

Step 9. Book a design consult

Bromic's design specialists review your photos, plans or drawings and recommend heaters, placement and controls, with coverage drawings, at no charge[2].

A complete quote has five lines: heaters and mounting kits; controls; installation; permits; running cost. US list prices on bromic.com (29 October 2026) start from $798 for a Tungsten Electric heater and $370 for an Affinity On/Off controller[413].

- [ ] Checklists from Steps 1–8 filled in
- [ ] Photos or plans ready
- [ ] Design consult booked

Covered outdoor kitchen and bar with flush ceiling heaters

Book a design consult and get a layout for your space.

NEXT STEP

Book a design consult



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Confirm every installation with your local building authority and a licensed electrician or gas fitter. Wi-Fi and app features need compatible Affinity Smart-Heat controls, sold separately. © 2026 Bromic Heating · bromic.com